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CS 499 4-2 Milestone Three Enhancement Two Algorithms and Data Structure

Artifact Description:

The artifact I selected for this milestone is my inventory application, which I am currently developing as part of my Computer Science Capstone project. This project is a full-stack application built using the MERN stack, including MongoDB, Express, React, and Node.js. As a reminder, this application is a web-based version of a previous Android app I developed using Kotlin for CS 360, where the data was stored locally on the device. In this new version, I am improving the overall design by implementing a more scalable backend and using a cloud-based database.

For this milestone, I focused on enhancing the backend by improving its structure and adding user-related functionality. I created a separate user model and implemented full CRUD operations for user data. Additionally, I reorganized the backend by separating routes, controllers, and models to improve code readability and maintainability. I also began implementing validation logic and security features, such as password hashing, to ensure that user data is handled properly. This milestone represents a finished backend product for this project. From now, I will only make changes to it as I see fit.

Justification for Inclusion:

I selected this artifact for my ePortfolio because it demonstrates my ability to apply algorithmic thinking and data structure concepts within a real-world application. One of the most important enhancements in this milestone was the implementation of validation logic in the user controller. This logic acts as a step-by-step validation algorithm that ensures all incoming user data meets specific requirements before being stored in the database.

The validation process includes checking that required fields are present, verifying that the email follows a valid format, enforcing password complexity rules, and ensuring that usernames meet length and character constraints. It also includes conditional logic for optional fields, such as validating a phone number only if it is provided. This demonstrates my ability to design logical algorithms that improve data integrity and prevent invalid data from entering the system.

In addition, this artifact also shows my understanding of data structures through the use of MongoDB collections and Mongoose schemas. By defining structured schemas for users and products, I am able to organize data efficiently and ensure consistency across the application. I also improved the overall design by modularizing the backend into separate components, which enhances scalability and makes the code easier to maintain.

Course Outcomes Reflection:

Through this enhancement, I made progress toward the course outcome of designing and evaluating computing solutions using algorithmic principles and data structures. The validation logic implemented in the user controller shows my ability to apply problem-solving techniques to real-world scenarios. It required me to think critically about different input cases and how to handle them effectively.

I also strengthened my understanding of data structures by working with MongoDB and Mongoose schemas to organize and manage application data. Additionally, I continued to apply best practices by improving the modular structure of the backend and incorporating security measures like password hashing using bcryptjs. These improvements show that I am making progress toward meeting the program outcomes related to both algorithmic thinking and software development practices.

At this stage, I believe I have met the goals I set for this enhancement, and I plan to continue improving the application by adding more advanced features such as search and filtering to further enhance efficiency.

Reflection on the Enhancement Process:

Throughout this enhancement process, I gained a deeper understanding of how algorithms and data structures are applied in backend development. One of the most valuable things I learned was how to design validation logic that handles multiple conditions in a clear and structured way. Writing these checks required careful planning to ensure that all possible input scenarios were handled correctly.

One of the challenges I faced was making sure that all validation rules worked together without conflicts, especially when dealing with optional fields like phone numbers. This taught me the importance of testing different edge cases.

I also learned the importance of organizing code in a modular way. Separating the backend into routes, controllers, and models made the application much easier to read and maintain. Additionally, implementing password hashing helped reinforce the importance of security when working with user data.

Overall, this milestone helped me connect theoretical concepts such as algorithms and data structures to practical application development. It also increased my confidence in building backend systems and reinforced the importance of writing clean, efficient, and secure code.